

Ladder or Satisfaction? Why Gallup and the World Values Survey Disagree on Income and Well-Being*

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September 30, 2026

Abstract

The two main sources of cross-national data on subjective well-being disagree: life evaluations are much more strongly related to GDP per capita in the Gallup World Poll (Cantril ladder, 0–10) than in the World Values Survey (WVS; life satisfaction, 1–10), here complemented with the European Values Study. In 187 country-years observed by both, log GDP per capita explains 0.53 of the variance of Gallup’s mean ladder but 0.16 of WVS mean satisfaction. Is this due to the question or to the samples? I randomize the wording (ladder vs. satisfaction) and the scale (0–10 vs. 1–10) in an online survey of 11,000 respondents in ten high-income countries. The ladder shifts answers

*Data and code: https://github.com/bixiou/wellbeing_gdp_region [TODO: public repository (e.g. OSF); Gallup data cannot be redistributed]. The survey was approved by the CIRED institutional review board (IRB-CIRED-2025-2) and pre-registered at osf.io/7mzn4. I thank Claudia Senik and participants in the 2024 STATEC Well-being Conference for helpful comments, Armon Rezai and the Vienna University of Economics and Business (WU Wien) for access to the Gallup World Poll data, and Julieta Toffoli for research assistance. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. Declaration of interest: none.

by -0.45 points, an effect that varies strongly across countries and fully accounts for Gallup's lower average level. However, wording has little effect on the relationship between income and well-being: across the ten countries, the income gradient is 4.35 with the ladder and 4.05 with the satisfaction question when both are asked to the same sample, close to Gallup's (4.20) and far from the WVS's (1.36). Within countries, the ladder does not steepen the income gradient either. Wording effects shift levels; sample differences drive the discrepancy in income gradients.

Keywords: subjective well-being; life satisfaction; Cantril ladder; question wording; survey mode; survey experiment.

JEL codes: I31; C83; C90.

1 Introduction

Cross-country comparisons of subjective well-being rely mostly on two surveys: the Gallup World Poll, which underlies the World Happiness Report, and the World Values Survey (WVS). They measure life evaluation with different questions. Gallup asks respondents to place themselves on a ladder whose top represents “the best possible life for you” (Cantril, 1965), on a scale from 0 to 10; the WVS (and the European Values Study, EVS, which asks the same question) asks “how satisfied are you with your life as a whole these days?”, on a scale from 1 to 10. They also differ in their samples: sampling frames, interview modes, questionnaire context, translations, and the years in which each country is surveyed.

The two surveys paint different pictures of the relationship between income and well-being. Deaton (2008) documented that Gallup's ladder is strongly related to GDP per capita, more so than WVS measures. The difference is large: in the 187 country-years (99 countries) observed by both surveys the same year, log GDP per capita explains 0.53 of the variance of Gallup's mean ladder, but only 0.16 of WVS mean satisfaction. Conclusions on whether “money buys happiness” across countries, or on whether a country's world region predicts well-being better than its income, depend on which survey one uses. [TODO: cite companion paper on region vs. income.]

Which difference is responsible: the question, or the samples? If the question,

then the two surveys measure different constructs—for instance, if the ladder elicits thoughts about wealth and status (Nilsson et al., 2024) or evaluative rather than hedonic judgments (Kahneman & Deaton, 2010)—and multiple indicators should be used. If the samples, then the surveys measure the same construct with different errors, and one should favor the dataset with the most reliable and comparable samples. Observational comparisons cannot answer this question, because the question and the sample always differ together (Bjørnskov, 2010).

I answer it with a survey experiment. In an online survey of 11,000 respondents in ten high-income countries (Fabre, 2025), I randomly assigned respondents to one of four versions of the life-evaluation question, crossing the wording (Gallup’s ladder vs. WVS life satisfaction) with the scale (0–10 vs. 1–10). The experiment estimates the question effect holding the sample constant. Combining it with Gallup and WVS data for the same countries, I decompose the observed Gallup–WVS gap into a question effect and a residual that captures differences between samples in a broad sense.

The results are clear-cut. The question matters for levels: the ladder shifts answers by -0.45 points on average relative to the satisfaction question, and the effect varies across countries, from -1.12 points in Spain to 0.36 in Japan. This question effect more than accounts for the lower average level of Gallup data in these countries. But it explains little of which countries score relatively higher in Gallup than in the WVS, and, crucially, it cannot account for the difference in income gradients. Across the ten countries, the slope of mean life evaluation on $\log_{10}$ GDP per capita is 4.20 in Gallup and 1.36 in the WVS; when the same sample answers either question, it is 4.35 with the ladder and 4.05 with the satisfaction question. Within countries too, the ladder does not steepen the income gradient. Direct comparisons of samples asked the same question confirm that sample and mode effects are as large as the question effect. The experiment covers only high-income countries, so it cannot exclude a larger wording effect in poorer countries, where the gap between the two surveys is largest; but nothing in the evidence points in this direction.

The paper contributes to the methodological literature on the measurement of subjective well-being (Kahneman & Krueger, 2006; Kaiser & Oswald, 2022; OECD, 2013; Stone & Mackie, 2013), in particular on question and scale effects (Bond & Lang, 2019; Conti & Pudney, 2011), context effects (Deaton, 2012; Deaton

& Stone, 2016), and mode effects (Dolan & Kavetsos, 2016), and to the debate on the cross-country relation between income and well-being (Deaton, 2008; Sacks et al., 2012; Stevenson & Wolfers, 2008).

2 Data

2.1 World Values Survey

The World Values Survey (WVS) is the longest-running cross-national survey of values and subjective well-being. I use the WVS time-series file (Haerpfer et al., 2022), which pools the seven waves conducted between 1981 and 2022, and complete it with the Joint EVS/WVS 2017–2022 dataset (EVS/WVS, 2024), which adds the 36 national surveys of the 2017 wave of the European Values Study (EVS, conducted in 2017–2021) and 2 recent WVS surveys (India 2023, Uzbekistan 2022). The EVS asks the same well-being questions, and the combination of the two surveys is standard practice (the so-called Integrated Values Surveys). The EVS and the WVS are distinct surveys: when both surveyed the same country the same year (3 cases), I keep both surveys as separate observations. For brevity, I refer to the combined data as the WVS: 501,911 respondents in 342 country-years and 112 countries. National samples are designed to be representative of the adult population, typically with 1,000 to 3,000 respondents, and most interviews are conducted face-to-face (though some recent national surveys used web, mail or mixed modes, e.g. the WVS in the United States in 2017 and in Japan in 2019, or the EVS in Switzerland in 2017). All statistics use the survey weights provided by the WVS and the EVS.

The WVS contains two questions on subjective well-being:

- *Happiness*: “Taking all things together, would you say you are: *Very happy*; *Quite happy*; *Not very happy*; *Not at all happy*.”
- *Life satisfaction*: “All things considered, how satisfied are you with your life as a whole these days?”, answered on a scale from 1 (*Completely dissatisfied*) to 10 (*Completely satisfied*).

Non-responses are rare (1.4% for satisfaction and 1.7% for happiness on average) and uncorrelated with GDP per capita (correlation: -0.09); they are excluded.

2.2 Gallup World Poll and World Happiness Report

The Gallup World Poll interviews around 1,000 adults per country and year in more than 140 countries, by telephone where coverage is high (typically in high-income countries) and face-to-face elsewhere (Gallup, 2022). Its life-evaluation question is the Cantril ladder (Cantril, 1965): “Please imagine a ladder, with steps numbered from 0 at the bottom to 10 at the top. The top of the ladder represents the best possible life for you and the bottom of the ladder represents the worst possible life for you. On which step of the ladder would you say you personally feel you stand at this time?”

I use the distribution of answers by country and wave, tabulated from the Gallup World Poll microdata (2,351 country-years, 168 countries, 2006–2023), from which I compute the same satisfaction indicators as for the WVS: the mean and the shares answering 6–10, 8–10, 9–10, 10, 0–4 and 0–2. These distributions are unweighted. To cover the most recent years, I also use the ladder means published in the World Happiness Report 2026 (World Happiness Report, 2026), which average Gallup data over three years up to 2023–2025.

2.3 Income

My benchmark income measure is the log of GDP per capita at purchasing power parity (PPP, constant 2021 international dollars) from the World Development Indicators (World Bank, 2026). As an alternative, I use GDP per capita at market exchange rates (constant 2015 US dollars). Missing values are filled in automatically, in three steps: (i) PPP series, which start in 1990, are extended backwards (and gaps filled) using the growth rate of real GDP per capita; (ii) for countries absent from the World Bank data (e.g. Taiwan, or Venezuela in recent years), I use the IMF World Economic Outlook (International Monetary Fund, 2026), whose values are in current dollars; I convert them into the constant dollars of the World Bank series by multiplying them by the ratio, in the same year, between the World Bank and the IMF values for the United States (a U.S. price deflator, since PPP and market-exchange-rate values coincide for the United States); (iii) remaining gaps are filled with the closest year available (Montenegro 1996). Territories without national accounts use the GDP of the closest entity (e.g. the United Kingdom for

Northern Ireland). A robustness check excludes all imputed values.

2.4 A survey experiment on question wording and scale

Gallup and the WVS differ in many respects. Gallup asks the Cantril ladder on a 0–10 scale, the WVS life satisfaction on a 1–10 scale. Gallup relies on a centralized, standardized design, with about 1,000 respondents per country every year, interviewed by telephone in high-income countries and face-to-face elsewhere; the WVS is fielded every five years or so by national teams, mostly face-to-face, with sampling designs, modes and survey years that vary across countries. These differences fall into two categories that could explain why the two datasets disagree: the question (wording and scale) and everything else—sampling frames, interview modes, questionnaire context, translations, and survey years. To separate the two, I embedded a randomized experiment in an original survey (Fabre, 2025). The survey was conducted online between April 15 and July 3, 2025, by Bilendi (Kantar in Saudi Arabia) on 11,000 respondents in 10 high-income countries: the United States (3,000), Japan (2,000), Saudi Arabia (1,000), Germany (1,048), the United Kingdom (826), France (798), Italy (756), Spain (603), Poland (500) and Switzerland (469). (The survey also covered Russia, where the well-being question was not asked.) Samples were stratified with quotas on gender, age, income, education, region and urbanicity (race in the U.S.; citizenship in Saudi Arabia), and results are reweighted to match population frequencies along these dimensions (weights trimmed between 0.25 and 4). Respondents failing an attention test or completing the questionnaire in less than 6 minutes were excluded. The project was approved by the CIRED institutional review board (IRB-CIRED-2025-2) and pre-registered at the Open Science Framework (osf.io/7mzn4).

Toward the end of the questionnaire, after questions on global policies, each respondent was randomly assigned to one of four versions of the life-evaluation question:

- *Ladder 0–10* (Gallup’s original): the Cantril ladder question quoted in Section 2.2, with steps numbered from 0 (*Worst possible*) to 10 (*Best possible*);
- *Ladder 1–10*: same wording, with steps numbered from 1 to 10;

- *Satisfaction 0–10*: the WVS question quoted in Section 2.1, with answers from 0 (*Completely dissatisfied*) to 10 (*Completely satisfied*);
- *Satisfaction 1–10* (WVS’s original): same wording, with answers from 1 to 10.

This 2×2 design identifies the effect of wording (ladder vs. satisfaction) and scale (0–10 vs. 1–10), and their interaction, holding the sample, mode, context and time constant. Within each country, the four branches are balanced (each receives between 21% and 30% of respondents).

3 Results

3.1 Experimental effects of wording and scale

Table 1 reports regressions of the answer on the randomized treatments, with country fixed effects, survey weights and heteroskedasticity-robust standard errors. The ladder wording shifts answers by -0.45 points on average (s.e. 0.05), i.e. respondents rate their lives lower on the “best possible life” ladder than when asked how satisfied they are. This is in line with evidence that the two questions elicit different considerations: in open-ended answers, the ladder evokes power and wealth more than life satisfaction does (Nilsson et al., 2024). The scale effect is smaller: answers on the 1–10 scale are 0.14 points higher than on the 0–10 scale (s.e. 0.05), which, once the 1–10 answers are linearly stretched to 0–10, amounts to -0.26 points. Wording and scale do not interact. Measured as the share of respondents answering 6 or more, the ladder changes the share “satisfied” by -9 percentage points.

Two further results matter for the interpretation of cross-country comparisons. First, the ladder does not steepen the income gradient *within* countries: the interaction between ladder wording and the income quartile is -0.004 ($p = .919$), against a gradient of 0.69 points per quartile (column 5). The scale does not change this conclusion: in a model interacting income with both wording and scale (column 6), the interaction between income and wording is 0.009 ($p = .883$) and the triple interaction is -0.023 ($p = .786$). So, at least among individuals

Table 1: Effects of question wording and scale on life evaluation, Fabre (2025) survey experiment

Dependent Variables: Model:	Answer (1)	Answer (2)	Answer (0–10 equiv.) (3)	Satisfied (6+) (4)	Answer (0–10 equiv.) (5)	Answer (0–10 equiv.) (6)
<i>Variables</i>						
Ladder wording (Gallup)	-0.451*** (0.046)	-0.455*** (0.067)	-0.456*** (0.067)	-0.088*** (0.013)	-0.464*** (0.123)	-0.494** (0.172)
Scale 1–10	0.137** (0.046)	0.132 (0.068)	-0.260*** (0.072)	-0.008 (0.013)	-0.276*** (0.046)	-0.540** (0.182)
Ladder wording (Gallup) $\times$ Scale 1–10		0.008 (0.092)	-0.041 (0.097)	0.028 (0.019)		0.055 (0.247)
Income quartile (1–4)					0.689*** (0.031)	0.635*** (0.043)
Income quartile (1–4) $\times$ Ladder wording (Gallup)					-0.004 (0.042)	0.009 (0.058)
Income quartile (1–4) $\times$ Scale 1–10						0.106 (0.061)
Income quartile (1–4) $\times$ Ladder wording (Gallup) $\times$ Scale 1–10						-0.022 (0.083)
<i>Fixed-effects</i>						
Country	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	11,000	11,000	11,000	11,000	11,000	11,000
R ²	0.05681	0.05681	0.05942	0.04208	0.15976	0.16024

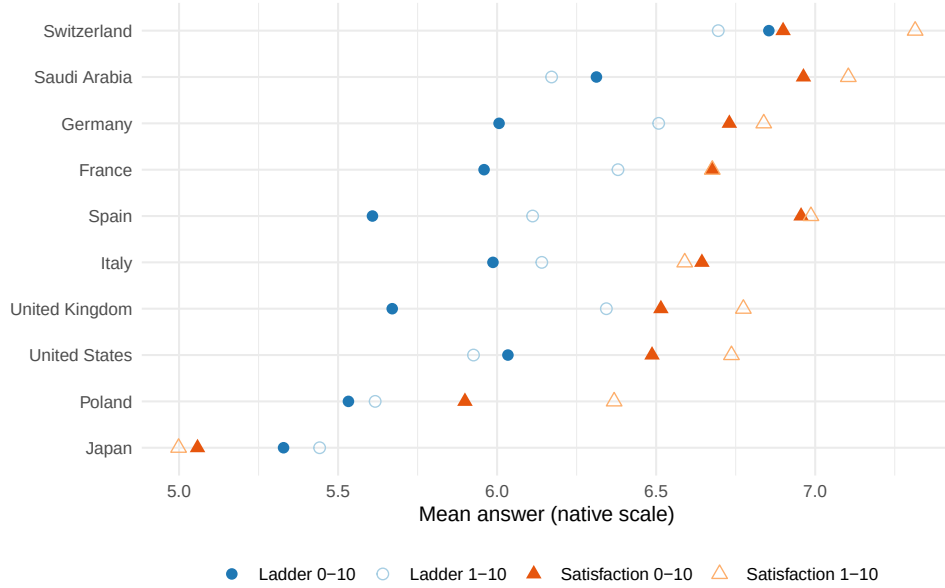
Heteroskedasticity-robust standard-errors in parentheses

*Signif. Codes: ***, 0.001, **, 0.01, *, 0.05*

Note: OLS with country fixed effects, survey weights (normalized within country) and heteroskedasticity-robust standard errors in parentheses. *Ladder wording:* Cantril ladder (vs. WVS life satisfaction). *Scale 1–10:* answers from 1 to 10 (vs. 0 to 10). In columns 3 and 5, 1–10 answers are linearly rescaled to 0–10. Column 5 interacts the wording with the respondent’s income quartile within their country. * $p < .05$, ** $p < .01$, *** $p < .001$.

in high-income countries, the ladder is not more “about money” than the satisfaction question. Second, the wording effect differs markedly across countries (Figure 1). It ranges from -1.12 points (Spain) to 0.36 (Japan), where the ladder yields *higher* answers than the satisfaction question, and the hypothesis of a common effect across countries is strongly rejected ($\chi^2(9) = 93.4$, $p < .001$). Question wording therefore affects not only the level but also the ranking of countries.

Figure 1: Mean answer by question variant and country, Fabre (2025)



Note: Weighted mean answer on the native scale of each variant (0–10 or 1–10).

3.2 Decomposing the Gallup–WVS gap in ten countries

For each country c of the experiment, I define the observed gap between Gallup and the WVS as $D_c = G_c - W_c$, where W_c is the mean satisfaction in the latest WVS or EVS survey of the country (from 2017–2022, except for Saudi Arabia, 2003) and G_c the mean ladder in Gallup the same year (or the closest year, at most three years apart; 9 countries have an exact match). Comparing the two datasets in the same year removes the influence of changes in well-being over time. The experiment measures, within the same sample, the *question effect* $Q_c = F_c(\text{ladder } 0\text{--}10) - F_c(\text{satisfaction } 1\text{--}10)$, where $F_c(v)$ is the mean answer to

variant v in country c . The residual $R_c = D_c - Q_c$ is the part of the gap due to everything that differs between the two datasets except the question: sampling frames, interview modes, questionnaire context, translations, and survey-specific time effects. This decomposition assumes that the question effect measured on-line in 2025 applies to the Gallup and WVS samples of earlier years.

Table 2 and Figure 2 present the decomposition for each country, and Table 3 summarizes it with 95% confidence intervals from a bootstrap that resamples respondents in the three surveys. On average, the gap between Gallup and WVS answers is $D = -0.60$ points. The question effect is even larger in absolute value: $Q = -0.71$ points. Wording (and scale) thus fully account for the fact that Gallup yields lower levels than the WVS in high-income countries, and the residual is small and not significantly different from zero ($R = 0.11$, 95% CI $[-0.04; 0.26]$). Since the two components move in opposite directions, another way to summarize the level difference is to compare the size of the two movements: the question accounts for $|\bar{Q}|/(|\bar{Q}| + |\bar{R}|) = 0.86$ (95% CI $[0.76; 0.99]$) of the gross difference in levels between Gallup and the WVS, and for 0.69 on average when this share is computed country by country. By this metric, wording is the main driver of the difference in *levels*.

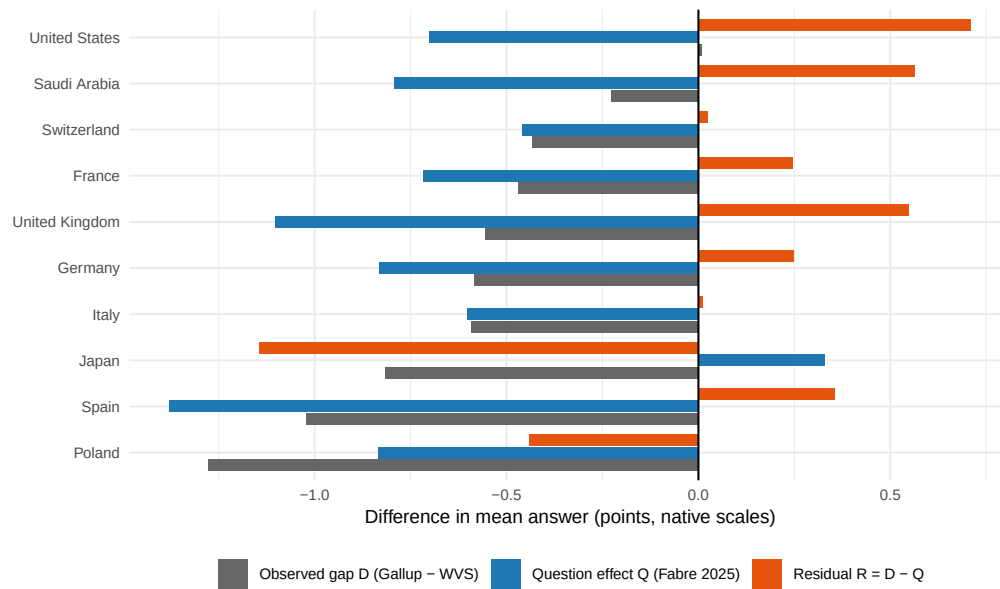
Table 2: Decomposition of the Gallup–WVS gap by country

Country	Year WVS/EVS (Gallup)	Survey: mode	Gallup (1)	WVS/EVS (2)	Gap D (1)–(2)	Ladder 0–10 Fabre (2025)	Satisf. 1–10	Question effect Q	Residual R $D - Q$
Poland	2017	EVS: CAPI	6.28	7.56	–1.28	5.53	6.37	–0.84	–0.44
Spain	2017	EVS: CAPI	6.47	7.49	–1.02	5.61	6.99	–1.38	0.36
Japan	2019	WVS: Mail	5.94	6.76	–0.82	5.33	5.00	0.33	–1.15
Italy	2018	EVS: CAPI	6.69	7.28	–0.59	5.99	6.59	–0.60	0.01
Germany	2018	WVS: CAPI	7.15	7.74	–0.58	6.01	6.84	–0.83	0.25
United Kingdom	2022	WVS: CAPI/Web/Mail	6.76	7.32	–0.56	5.67	6.77	–1.10	0.55
France	2018	EVS: CAPI	6.79	7.26	–0.47	5.96	6.68	–0.72	0.25
Switzerland	2017	EVS: Web/Mail/CAPI	7.58	8.02	–0.43	6.85	7.31	–0.46	0.03
Saudi Arabia	2003 (2006)	WVS: Face-to-face	7.06	7.28	–0.23	6.31	7.10	–0.79	0.57
United States	2017	WVS: Web/Phone	7.23	7.22	0.01	6.03	6.74	–0.70	0.71

Note: Mean answers on native scales (Gallup: 0–10; WVS/EVS: 1–10). The Gallup year is given in parentheses when it differs from the WVS year. Survey: mode: source of the WVS/EVS data and main interview mode(s) (CAPI: computer-assisted personal interviews).

What matters for correlations with GDP and for country rankings, however, is not the level of the gap but its variation across countries, which the previous metrics do not capture: a question effect that shifted all countries by the same amount would score high on them while leaving correlations and rankings unchanged. Here, the question effect explains little: the share of the cross-country variance of the gap attributable to the question, $\text{cov}(D, Q)/\text{var}(D)$, is 0.11 (95%

Figure 2: Observed Gallup–WVS gap, question effect and residual by country



CI $[-0.33; 0.55]$), and the residual accounts for the rest. Predicting a country's Gallup score by adding its experimentally measured question effect to its WVS score does worse (root mean squared error: 0.54) than simply adding the average gap (0.35). The conclusion is the same when excluding Saudi Arabia, the only country whose latest WVS survey is not recent (variance share: 0.16), and for the share of respondents answering 6 or more (0.04). For instance, Japan is the only country with a positive question effect, but the gap is negative there; and in the United States, the gap is nil although the question effect is as large as in other countries.

Two direct comparisons of samples asked the same question corroborate the importance of the residual. First, Gallup respondents (World Happiness Report, 2023–2025) rate their lives 0.75 points higher on the 0–10 ladder than Fabre (2025) online respondents asked the same question, with differences ranging from 0.16 to 1.24 across countries. Second, WVS respondents report 0.75 points higher satisfaction than online respondents asked the WVS question (this comparison also includes changes over time, as WVS surveys are older). These sample (and mode) effects are of the same order of magnitude as the question effect, and they vary across countries. Even in the United States, where the 2017 WVS was itself con-

Table 3: Decomposition of the Gallup–WVS gap: summary

Statistic	Estimate [95% bootstrap CI]
Mean gap D (Gallup – WVS)	–0.60 [–0.64; –0.55]
Mean question effect Q (wording + scale)	–0.71 [–0.84; –0.56]
Mean residual R (sampling, mode, context)	0.11 [–0.04; 0.26]
Share of mean gap due to question ($\bar{Q}/\bar{D}$)	1.19 [0.94; 1.45]
Share of gross level difference due to question ($ \bar{Q} /(\bar{Q} + \bar{R})$)	0.86 [0.76; 0.99]
Same, country average ($ \bar{Q}_c /(\bar{Q}_c + \bar{R}_c)$)	0.69 [0.56; 0.72]
Share of cross-country variance of D due to Q ($\text{cov}(D, Q)/\text{var}(D)$)	0.11 [–0.33; 0.55]
Share of cross-country variance of D due to R	0.89 [0.45; 1.33]
Mean absolute gap $ \bar{D} $	0.60 [0.56; 0.65]
Mean absolute residual $ \bar{R} $	0.43 [0.33; 0.60]
Correlation between D and Q	0.09 [–0.26; 0.41]
Mean gap, share satisfied (6+)	–0.07 [–0.08; –0.06]
Mean question effect, share satisfied (6+)	–0.11 [–0.14; –0.08]
Variance share due to Q , share satisfied	0.04 [–0.33; 0.45]

Note: 10 countries. 95% confidence intervals from 1,000 bootstrap replications resampling respondents within country $\times$ variant (Fabre), within country (WVS), and answers within country (Gallup, multinomial).

ducted online, WVS respondents report 0.49 points higher satisfaction than Fabre (2025) respondents.

3.3 Income gradients in the ten countries: past vs. new data

The decomposition above concerns levels. The question that motivates this paper, however, is whether the two datasets disagree on the *relationship* between income and well-being because of their questions. The ten countries of the experiment provide a direct test (Table 4). In the past data, the income gradient is much steeper in Gallup than in the WVS: across the ten countries, the slope of mean life evaluation on $\log_{10}$ GDP per capita is 4.20 in Gallup and 1.36 in the WVS, a difference of 2.84 (95% bootstrap CI [2.37; 3.26]), and log GDP explains 0.75 of the cross-country variance in Gallup against 0.16 in the WVS. In the new data, where the same sample answers either question, the gradient is similar with both wordings: 4.35 with the ladder and 4.05 with the satisfaction question, a difference of 0.30 (95% CI [–1.21; 1.96]); the two other variants yield intermediate slopes. Both original questions thus yield a gradient close to Gallup’s. The effect of wording on the gradient is imprecisely estimated, but it is significantly smaller

than the difference between Gallup and the WVS: the part of this difference that is not due to the question is 2.54 (95% CI [0.90; 4.18]). The flat gradient of the WVS is thus mostly specific to its samples, not to its question. This is consistent with the individual-level evidence that the ladder does not steepen the income gradient within countries (Section 3.1). Two caveats apply. First, the bootstrap confidence intervals reflect the sampling of respondents and treat the ten countries as given; with ten countries, the slopes themselves are imprecise (Table 4 reports their standard errors across countries). Second, the R^2 differ more than the slopes, because the satisfaction question produces more dispersion around the income gradient (notably in Japan), and the decompositions between income and region are uninformative here, since Poland is the only Eastern European country.

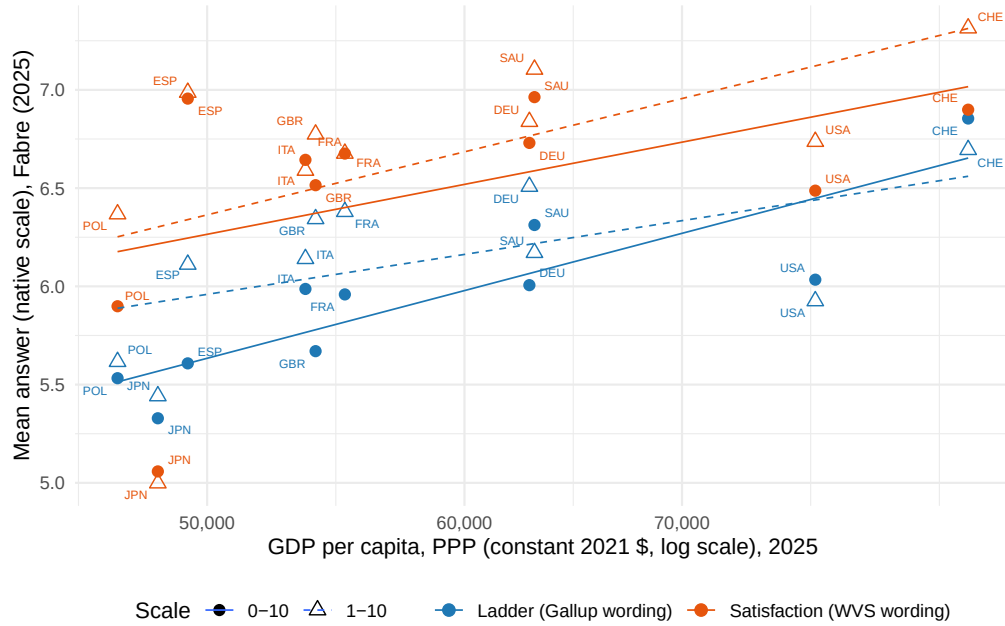
Table 4: Income gradient of national well-being in the ten countries of the experiment: past vs. new data

Data	Slope on $\log_{10}$ GDP	s.e.	R^2 income	R^2 region	Share due to income
Gallup (ladder 0–10)	4.20	0.80	0.75	0.29	0.78
WVS/EVS (satisfaction 1–10)	1.36	1.44	0.16	0.34	0.35
Fabre 2025: ladder 0–10	4.35	1.05	0.76	0.14	0.91
Fabre 2025: ladder 1–10	2.57	1.36	0.33	0.50	0.36
Fabre 2025: satisfaction 0–10	3.21	2.03	0.23	0.36	0.35
Fabre 2025: satisfaction 1–10	4.05	2.09	0.31	0.29	0.52

Note: OLS across the 10 countries of the experiment. Gallup and WVS/EVS: latest WVS/EVS survey and Gallup the same year, GDP of that year. Fabre (2025): mean answer by variant, GDP of 2025. Standard errors robust to heteroskedasticity (HC1). R^2 income and R^2 region: R^2 of the regressions on log GDP and on region dummies (Western, Asia, Eastern Europe); share due to income: share of the R^2 of the regression on both attributed to income by the Shapley (LMG) decomposition.

Two further comparisons of samples asked the same question confirm that sample differences are large. First, the EVS and the WVS surveyed 10 countries separately during 2017–2022, with the same satisfaction question; their mean satisfaction differs by 0.28 points on average (up to 0.53), a magnitude comparable to the cross-country variation of the Gallup–WVS gap. Second, as noted above, Gallup and WVS respondents report markedly higher well-being than online respondents asked the same question.

Figure 3: Mean life evaluation vs. GDP per capita (PPP), by question variant, Fabre (2025)



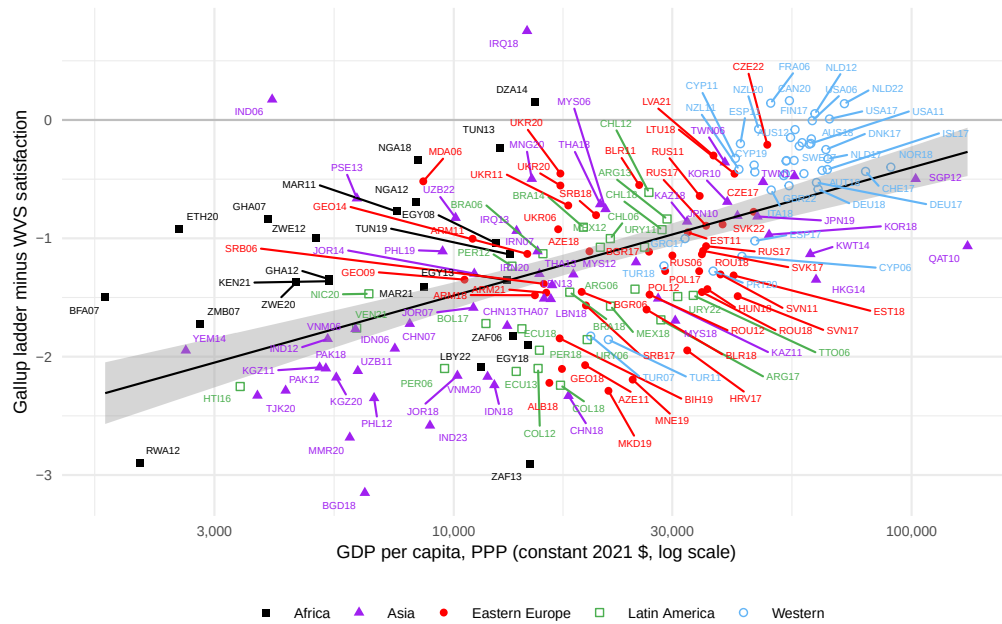
Note: Mean answer on the native scale of each variant (0–10 or 1–10), ten countries surveyed in 2025. Colors: wording; shapes and line types: scale. Lines: OLS fits by variant.

3.4 The global discrepancy

The ten countries of the experiment are all high-income, whereas the Gallup–WVS discrepancy matters most for the global relation between GDP and well-being. Restricting both datasets to the 187 country-years (99 countries) observed by both surveys the same year, log GDP explains 0.53 of the variance of Gallup’s mean ladder but only 0.16 of WVS mean satisfaction, and income accounts for 51% of the variance explained jointly with region in Gallup against 24% in the WVS. The gap between the two datasets increases strongly with GDP (Figure 4): the slope is 1.08 per unit of $\log_{10}$ GDP (s.e. 0.14, clustered by country), with WVS answers exceeding Gallup answers by around two points in low-income countries, against less than half a point in high-income countries.

Could the question explain this pattern? Within the experiment, the question effect is not significantly related to GDP (slope 0.30, s.e. 1.75), which is unsurprising given the narrow range of GDP among the ten countries. Even taking this point estimate at face value and extrapolating it to all countries, adjusting WVS

Figure 4: Gallup ladder minus WVS satisfaction in the same country-year, by GDP per capita



Note: Mean answers on native scales (Gallup: 0–10; WVS: 1–10), 187 country-years observed by both surveys. Line: OLS fit with 95% confidence band.

answers for the predicted question effect would raise the R^2 of WVS satisfaction on log GDP only from 0.16 to 0.26, far from Gallup's 0.53. The experiment cannot rule out that the wording effect is much larger in low-income countries—for instance if the “best possible life” evokes material comparisons more strongly where material deprivation is widespread—but nothing in the high-income evidence points in that direction: the ladder does not steepen the individual income gradient (Section 3.1), and the country-level question effects are unrelated to the observed gaps.

3.5 Which dataset has more representative samples?

Since the discrepancy stems from samples rather than questions, which dataset is closer to the truth? The methodological documentation offers arguments on both sides. Gallup applies a centralized, standardized design in all countries: probability samples of about 1,000 adults per year, drawn by random-digit dialing where telephone coverage is high and by multistage area sampling elsewhere, with random selection of the respondent within the household, post-stratification weights on gender, age and, where possible, education, and documented coverage exclusions (Gallup, 2023). But its samples are small; its face-to-face samples select households by random route, with replacement of non-responding households; response rates are not published; interviews switched from face-to-face to telephone during the pandemic (2020–2021); and in 2023, a fifth of the interviews in 27 high-income countries came from web panels, most of them opt-in (Gallup, 2023). Blanchflower (2024) also questions the reliability of Gallup estimates for subgroups within countries. The WVS requires probability samples of at least 1,200 respondents but allows quota selection at the last stage, and, being fielded by national teams, its sampling designs, modes and survey years vary across countries (World Values Survey Association, 2023). Automated tests suggesting duplicated interviews in international survey programs (Kuriakose & Robbins, 2016) have been disputed (Pew Research Center, 2016) and cannot be applied to Gallup, whose microdata are not public.

I test the representativeness of WVS/EVS samples directly (Table 5). Compared with population benchmarks (Barro & Lee, 2013; World Bank, 2026), WVS samples over-represent tertiary-educated adults (aged 25 and over) by a median

factor of 1.98 in countries with a GDP per capita below 10,000 dollars, against 1.05 in countries above 30,000 dollars (correlation of the log ratio with log GDP: -0.40 , 295 surveys). They also over-represent urban residents in poorer countries (median ratio 2.19 vs. 1.22, 76 recent surveys) and under-represent people aged 65 or more (0.64 vs. 1.01). The internal criterion of Kohler (2007)—women should make up half of married or cohabiting respondents—is violated at the 5% level in 36% of surveys (42% in countries below 10,000 dollars and 34% above 30,000 dollars), with a median deviation of 1.1 percentage points. WVS samples are thus less representative in poorer countries, which is where their gap with Gallup is largest. However, these observable imbalances do not explain the gap. Controlling for log GDP, the over-representation of the tertiary-educated does not predict the Gallup–WVS gap (coefficient 0.07, s.e. 0.07, $p = .311$, 174 surveys), and reweighting each WVS sample to the benchmark share of tertiary-educated adults barely changes its income gradient: in the country-years also covered by Gallup, the R^2 of mean satisfaction on log GDP goes from 0.19 to 0.20, against 0.54 for Gallup.

Table 5: Representativeness of WVS/EVS samples: sample shares relative to population benchmarks

Measure	Surveys	Median				Correlation with log GDP
		All	GDP < 10k	10–30k	> 30k	
Tertiary education, 25+ (ratio sample/benchmark)	295	1.18	1.98	1.18	1.05	-0.40
Urban (ratio sample/benchmark)	76	1.44	2.19	1.45	1.22	-0.69
Aged 65+ (ratio sample/benchmark)	335	0.90	0.64	0.88	1.01	0.33
Women among married respondents – 0.5	337	0.01	-0.00	0.01	0.01	0.19

Note: One observation per WVS or EVS survey. Ratios: weighted sample share divided by the population share, medians across surveys (a ratio above 1 indicates over-representation). Benchmarks (World Bank WDI): share of the population aged 25+ with at least short-cycle tertiary education (UNESCO), or with complete or incomplete tertiary education (Barro & Lee, 2013) before 2011; urban population share (national definitions; sample: urban and peri-urban respondents, waves 4, 5 and 7 only); share aged 65+ among the population aged 15+ (sample: 65+ among respondents, usually 18+). Women among married: unweighted share of women among married or cohabiting respondents minus 0.5. Correlations with log GDP per capita (PPP) of the log ratios (or of the deviation).

Overall, three arguments favor Gallup for cross-country comparisons: its standardized design and weighting, the lower representativeness of WVS samples in poorer countries, and the fact that an independent online sample reproduces Gallup’s income gradient in high-income countries (Section 3.3). But the discrepancy cannot be traced to the observable composition of WVS samples, which suggests that it stems from unobserved differences in who responds, how, and

in which context; and Gallup’s recent changes of interview modes call for caution when comparing years. (The Gallup distributions used in this paper are unweighted, but their three-year averages correlate at 0.993 with the weighted averages published in the World Happiness Report.)

4 Discussion

The experiment shows that the question matters for levels: the Cantril ladder yields lower answers than the WVS life-satisfaction question, and this difference fully accounts for the lower average level of Gallup data in high-income countries. But the question does not explain the cross-country pattern of the Gallup–WVS gap, which is what drives differences in correlations with GDP and in country rankings: asked to the same sample, both questions yield similar income gradients, close to Gallup’s. The pattern is explained by the residual, i.e. by differences between samples in the broad sense: sampling frames, interview modes (Conti & Pudney, 2011; Dolan & Kavetsos, 2016), questionnaire context (Deaton, 2012; Deaton & Stone, 2016), translations, and time.

Several limitations should be kept in mind. First, the experiment covers ten high-income countries, whereas the gap varies most across income levels. The question effect could be different in low- and middle-income countries, where both Gallup and the WVS mostly interview face-to-face; only an experiment in these countries can settle this. Second, the residual lumps together several factors that the design cannot separate. In particular, the question effect was measured with the survey’s own translations, toward the end of a questionnaire on global policies, and online; it may differ from the effect that would be obtained with official Gallup and WVS translations, at the beginning of an interview, and face-to-face or by phone. Third, the decomposition assumes that the question effect measured in 2025 applies to the WVS/EVS and Gallup surveys of 2017–2022 (2003 for Saudi Arabia). Fourth, with ten countries, cross-country statistics are imprecise, as shown by the bootstrap confidence intervals. Finally, online panel respondents report markedly lower well-being than Gallup and WVS respondents asked the same question, a sample and mode effect that deserves attention in its own right as online panels become common in well-being research.

5 Conclusion

The Gallup World Poll and the World Values Survey disagree on how strongly national well-being is related to income. A survey experiment in ten high-income countries shows that their different questions explain why Gallup's life evaluations are lower on average, but not why the two surveys disagree on the income gradient: asked to the same sample, the ladder and the satisfaction question yield similar gradients, close to Gallup's. The discrepancy therefore comes mainly from differences in samples, interview modes, and context, rather than from the constructs measured.

This has two implications. First, the Cantril ladder and the life-satisfaction question can be treated as measures of the same construct up to a shift in level, at least in high-income countries; the choice between datasets should be based on the quality and comparability of their samples. Gallup's standardized annual methodology is an argument in its favor for cross-country comparisons, but the two surveys disagree most in low-income countries, where there is not yet evidence on which is closer to the truth. Second, the country-specific wording effects imply that rankings of countries are sensitive to the question asked, even among high-income countries. Replicating the experiment face-to-face in low- and middle-income countries, with official translations and at the beginning of the interview, is the natural next step.

Declaration of generative AI and AI-assisted technologies in the writing process

[TODO: During the preparation of this work the author used Claude (Anthropic) to write analysis code and draft parts of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. (Adapt to actual use.)]

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